MEDI-SCRIPT-AI: A Hybrid Detect–Read–Verify Pipeline for Digitising Handwritten Medical Prescriptions into Structured Records

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***Abstract*—Handwritten prescriptions remain a leading source of dispensing errors, yet existing digitisation systems either classify isolated characters, rely on large proprietary corpora, or treat the page as a single OCR problem. We present MEDI-SCRIPT-AI, a modular pipeline that separates *where* information is written from *what* it says and then *verifies* the answer against medical knowledge. A fine-tuned YOLOv8-nano detector localises handwritten medicine lines; each line is enhanced and read by a three-tier recogniser (a multimodal vision-language model, falling back to a fine-tuned TrOCR-Large whose N-best list is rescored by a character-level domain language model); the text is corrected against a 384,604-entry medicine lexicon with TF-IDF character n-gram retrieval; and all remaining printed or demographic text is recovered with full-page EasyOCR, regular expressions and a spatial anchor parser. A constrained LLM stage re-assigns unclassified fragments and expands dosage abbreviations into patient-readable instructions. The detector reaches 86.1% mAP50 (46.2% mAP50–95, 81.9% precision, 80.6% recall) on a real handwritten-prescription benchmark, trains in about four minutes on one GPU and occupies 6.2 MB. Qualitative error analysis of an earlier confidence-gated configuration shows *safe failure*—an empty field rather than a fluent wrong drug—when recognition is uncertain, and also exposes the fluent-hallucination mode that motivates verification. We position the work against the recent CNN–MLP framework of Biswas *et al.* [1], enumerate which design limitations it removes, and state candidly which accuracy claims we cannot yet make.**

***Index Terms*—handwritten text recognition, medical prescription digitisation, YOLOv8, TrOCR, vision-language models, lexicon correction, electronic health records**

I. INTRODUCTION

MEDICAL prescriptions are the interface between a clinician's decision and the dispensed treatment. In much of South Asia they are still written by hand on pre-printed pads with irregular layouts, and illegible drug names or dosages are a recognised cause of preventable harm: the Institute of Medicine estimated that medication errors injure at least 1.5 million people per year in the United States, with the extra cost of drug-related injuries in hospitals alone at least $3.5 billion annually [2]. Manual transcription by pharmacists or data-entry staff is slow, error-prone and a barrier to electronic health record (EHR) integration.

A prescription is not an ordinary document. It is a *semantically structured* artefact: different regions carry different kinds of information (patient demographics, diagnosis, one medicine per line with dosage), the printed header is machine-readable while the body is free-form handwriting, and the vocabulary—thousands of brand names, many absent from general language models—is highly specialised. Any practical system must therefore answer two questions in sequence: *where* is each piece of information, and *what* does it say?

A. Problem statement

Given one RGB photograph P of a handwritten prescription, produce a structured record

$$R = \mathcal{F}(P) = \big(\text{doctor},\ \text{clinic},\ \text{date},\ \text{patient} = (\text{name}, \text{age}, \text{sex}),\ \mathcal{M},\ \text{other}\big), \tag{1}$$

where $\mathcal{M} = \{m_1, \ldots, m_k\}$ is the ordered list of prescribed medicine lines (name, strength and dosage notation such as `1-0-1`). The system must prefer an *abstention* (empty or flagged field) over a confidently wrong value, because in this domain a fluent error is more dangerous than an obvious omission.

B. Motivation from the base study

The study of Biswas *et al.* [1] (hereafter the *base paper*) is a comprehensive recent attempt at this task. It segments scanned prescriptions with contours, separates printed from handwritten patches with a CNN–SVM, and recognises each attribute (name, age, gender, symptom, medicine) with a CNN feature extractor followed by a multilayer perceptron trained on a very large private collection, exporting the result as CSV for EHR use. Its strengths are scale and an explicit end-to-end data path. Its design, however, has four limitations that motivated our work: (i) recognition is *closed-vocabulary classification of glyph or word classes* trained per attribute, so a drug name unseen at training time cannot be produced; (ii) there is no explicit knowledge-based verification of drug names; (iii) dosage notation and layout-dependent fields are not modelled; and (iv) its reported figures depend on a proprietary data set that cannot be re-used or inspected. MEDI-SCRIPT-AI is designed to address exactly these four points.

C. Contributions

1) A **detect–read–verify architecture** that pairs a lightweight object detector with line-level sequence recognition instead of per-attribute glyph classifiers (Section IV).

2) A **tiered recogniser**: multimodal LLM → fine-tuned TrOCR-Large with domain-LM rescoring → optional confidence-gated abstention, which keeps the system usable offline and exposes (rather than hides) hallucination.
3) A **large-lexicon correction layer** using TF-IDF character n-gram retrieval (one sparse matrix–vector product over a 384,604-entry medicine list per query), with dosage-aware tokenisation.
4) A **layout-aware parser** combining regular expressions with spatial anchor search for printed and demographic fields, including explicit fall-backs for common OCR confusions.
5) A **constrained LLM post-processing stage** that fills only empty fields and translates dosage abbreviations without giving medical advice.
6) An **incremental TrOCR training recipe** (v1→v4) with synthetic dosage data, and an open, reproducible code base (Section VI).
7) A **candid evaluation** that reports the measured detector metrics, documents failure modes, and separates claims we can support from those we cannot (Section VII).

II. Related Work

A. Classical and classifier-based prescription recognition

Early systems relied on fixed templates or keyword spotting with hidden Markov models over printed sections, and later on handcrafted features with SVM or naive-Bayes classifiers. The base paper [1] summarises this line and its limits: dependence on printed text, poor scalability and single-language scope. Digit and glyph recognisers trained on MNIST/EMNIST [3] give strong isolated-character accuracy but need an explicit segmentation of characters, which cursive medical handwriting makes unreliable.

B. CNN–MLP and CTC pipelines

Convolutional feature extractors followed by recurrent layers and connectionist temporal classification (CTC) [4], [5] are the standard recipe for line-level handwriting recognition. The base paper positions its model in this family but, in practice, uses CNN features with MLP classifiers over character and word classes. Classification heads bound the output space by construction, whereas sequence decoders can emit any string.

C. Transformer-based recognition

TrOCR [6] replaces the CNN/RNN stack with an image Transformer encoder [7] and an autoregressive text Transformer decoder [8], pre-trained on large synthetic and real text-line corpora. It reads a whole line without character segmentation, but its decoder carries a general-language prior and can hallucinate fluent but unrelated words on illegible handwriting—a failure we observe directly (Section VII). OCR-free document transformers such as Donut [9] and layout-aware models such as LayoutLMv3 [10] treat the page holistically; in our test they transferred poorly from synthetic to real prescriptions.

D. Detection-then-recognition

Object detectors [11], [12] localise regions that are then passed to a recogniser. This decomposition lets each model be trained on the data it needs and exposes a clean place to insert domain knowledge between stages; it is the basis of Medi-Script-AI.

E. Multimodal foundation models

Large vision-language models [13] read handwriting zero-shot and can follow format instructions, but are remote services, rate-limited and opaque. We use one as the *primary* reader while retaining a local fine-tuned model as fall-back, and constrain its prompts to transcription and abbreviation expansion.

F. Gap

Prior work either classifies glyphs/words inside a closed vocabulary, or applies one generic reader to the whole page. None combines (a) a learned line detector, (b) open-vocabulary sequence recognition, (c) large-lexicon verification with abstention, and (d) layout-aware parsing of the non-handwritten fields in one reproducible, offline-capable system. Table I makes this comparison explicit.

III. Materials

A. Detection data

We use two public Roboflow Universe exports of handwritten prescriptions with bounding boxes around handwritten medicine lines (Table II). Dataset D2 is the Bannari Amman Institute handwritten-recognition set [17], on which the detector reported in Section VII was trained and evaluated; Dataset D1 is a larger Roboflow export of doctors' prescriptions used for pipeline development. Roboflow exports include augmented copies of source images (visible as duplicate image names with different hash suffixes), so image counts overstate the number of independent prescriptions.

B. Recognition data

Pseudo-labelled crops. YOLO crops of real prescriptions were transcribed with a multimodal LLM prompted to output exactly what is written, including abbreviations and dosage symbols, producing an image/`Text` table used for TrOCR v2 and v3. *Synthetic words.* For every entry of the lexicon, three renderings with random handwriting-style fonts (six cursive Google Fonts), colour, rotation and blur were generated at height 64 px (3,531 training images in the released manifest). *Synthetic dosage strings.* One thousand images of strings such as `1-0-1 x 5 days` or `1/2 - 0 - 1/2 for 1 week` were rendered the same way for the v4 increment. *RxHandBD.* A public set of Bangladeshi prescription word images was used for an initial frozen-encoder run.

C. Lexicons

The medicine lexicon holds 384,604 Indian brand names; a symptom lexicon holds 35 phrases. A curated 1,142-entry subset was used in the earliest experiments (Section VII). The full medicine list is not redistributed (size and licence); the repository documents how to rebuild it.

TABLE I: Design-level comparison with the base study. Reported accuracies of the two systems are *not* comparable (different data, private vs. public), and are therefore listed separately in Table VII.

Aspect	**Base paper [1]**	**MEDI-SCRIPT-AI (this work)**
Localisation	Contour detection + CNN–SVM printed/handwritten classifier	YOLOv8-nano learned line detector (6.2 MB); EasyOCR + rules for printed text
Recognition unit	Characters / words classified per attribute (CNN features + MLP)	Whole line via image-to-text Transformer (TrOCR-Large) or multimodal LLM
Vocabulary	Closed (trained classes; EMNIST 47 classes)	Open decoding, constrained afterwards by a 384,604-entry lexicon
Knowledge verification	None reported	TF-IDF char n-gram lexicon correction; domain LM rescoring; abstention gate
Dosage notation	Not modelled	Synthetic-data fine-tuning (v4) and dosage-aware tokenisation
Printed text	Tesseract [14]	EasyOCR (CRAFT + CRNN) [15], [16]
Layout fields (name, age, sex, date)	Separate per-attribute models on pre-cropped fields	Spatial anchor parser with regex fall-backs; works on free layouts
Post-processing	CSV export	LLM field re-assignment + dosage explanation; JSON / CSV
Training data	>400,000 proprietary prescriptions	357-image public detection set, synthetic and pseudo-labelled crops (data and code released)
Failure behaviour	Wrong class label	Empty field in the gated configuration; verification flags otherwise
Deployment	Notebook	FastAPI service, web UI, Docker image, CLI

TABLE II: Detection data sets (single class: handwritten prescription line).

	Train	**Valid**	**Test**	**Total**
D1 images	884	219	135	1,238
D1 boxes	3,976	972	556	5,504
D2 images (Bannari)	250	73	34	357
D2 boxes	740	216	90	1,046

D. Ethics and privacy

No data collected from patients are used. The data sets are public and de-identified at source. Because the LLM tier transmits crops to an external service, deployments handling real patient data should use the local-only configuration (Section VI).

IV. PROPOSED METHODOLOGY

Figure 1 gives the system overview and Fig. 2 the control flow of one inference. The pipeline has five stages organised into two parallel paths that are merged at the end.

A. Stage 1: handwritten-line localisation

We fine-tune **YOLOv8-nano** [12] (3.0 M parameters, 8.1 GFLOPs) as a single-class detector at 640×640 input. Detections with confidence ≥ 0.25 form the set $B = \{b_j\}$ of line boxes. The nano variant was chosen because it trains in minutes, produces a 6.2 MB checkpoint small enough to ship in the repository, and runs in milliseconds, which leaves the latency budget to the recognisers. Each box is padded by 5 px before cropping to avoid clipping ascenders and descenders of cursive strokes.

B. Stage 2a: crop enhancement

Handwritten crops are small, low-contrast and unevenly lit. Each crop C passes through the chain in Fig. 4: Lanczos upscaling by s=2, grayscale conversion $I_{\text{gray}} = 0.299R + 0.587G + 0.114B$, a 3×3 Gaussian blur, contrast-limited adaptive histogram equalisation (CLAHE) [18] with clip limit 2.5 on a 4×4 grid, adaptive Gaussian thresholding (block 15, constant 8), and a 2×2 dilation that reconnects thin strokes. Local rather than global thresholding is deliberate: global Otsu binarisation [19], used on the full page by the base paper, fails under uneven illumination, and a per-crop operation adapts to each line's lighting. A non-local-means denoiser was replaced by the Gaussian blur because it was $O(N^2)$ and stalled on large crops, while the blur costs under one millisecond.

C. Stage 2b: three-tier recognition

a) Tier 1: multimodal LLM.: The enhanced crop is sent to a Gemini vision model [13] with a prompt that casts it as a pharmacist reading *one* medicine line, forbids explanations, and requires the literal token `UNREADABLE` when the line is unclear or not a medicine. An `UNREADABLE` reply or any API error returns the empty string and triggers Tier 2.

b) Tier 2: fine-tuned TrOCR-Large.: The crop is resized to 384×384 and split into 16×16 patches; an encoder E produces visual tokens and an autoregressive decoder factorises

$$p(t \mid C) = \prod_{k=1}^{|t|} p\big(t_k \mid t_{<k}, E(C)\big). \tag{2}$$

Decoding uses beam search (N=2 returned hypotheses for medicine fields, `no_repeat_ngram_size`= 3, at most 24 new tokens). The confidence of hypothesis t_i is the mean token probability, $c_i = 100 \cdot \frac{1}{|t_i|}\sum_k p(t_{i,k} \mid \cdot)$. The newest available checkpoint (v3, v2, v1; the v4 dosage checkpoint is opt-in) is loaded, and the public `trocr-large-handwritten` model is the last resort.

c) Domain language-model rescoring.: Because a general decoder prefers fluent English, hypotheses are re-ranked with a

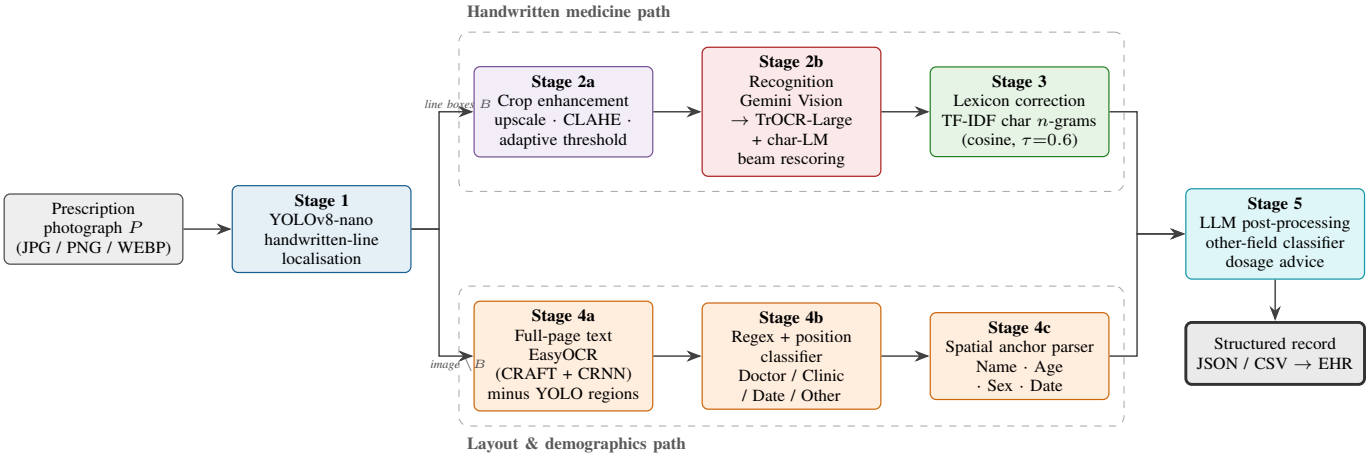


Fig. 1: MEDI-SCRIPT-AI architecture. A learned detector (Stage 1) splits the image into handwritten medicine lines (upper path: enhancement, three-tier recognition, lexicon correction) and everything else (lower path: full-page OCR, regex/position classification, spatial anchor parsing). A constrained LLM stage merges both into the final structured record.

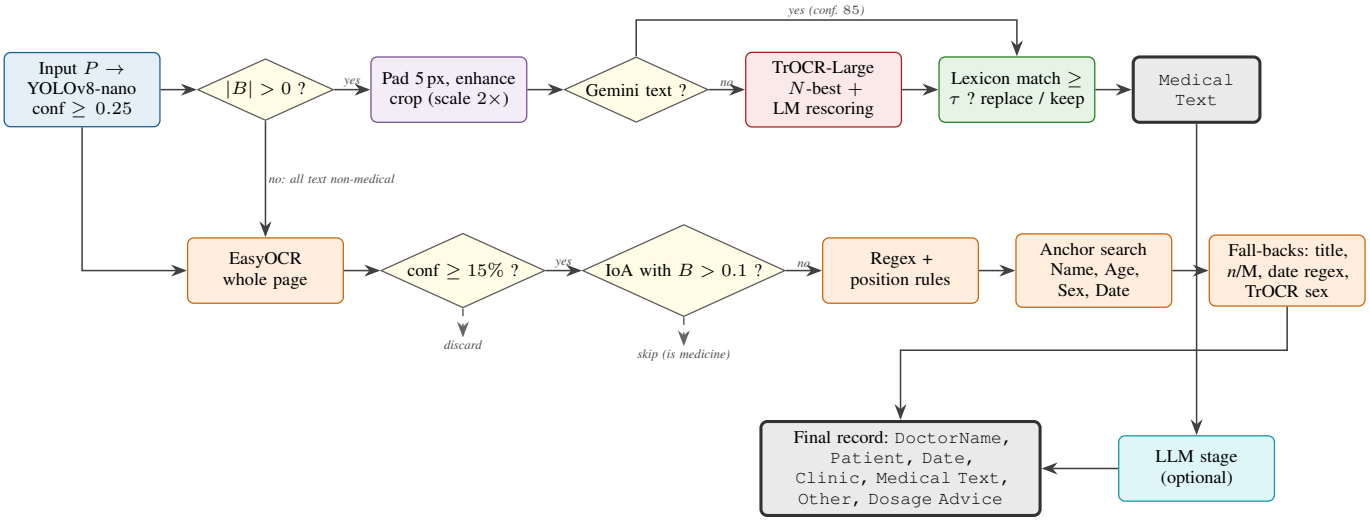


Fig. 2: Control flow of a single inference. Diamonds are decisions; dashed arrows denote discarded or skipped items. The Gemini branch is optional: without an API key every crop takes the local TrOCR route.

character trigram model trained on the lexicons ($|V|$ symbols, Laplace smoothing):

$$\mathrm{LM}(t) = \frac{1}{|t|} \sum_{j} \log \frac{n(g_j) + 1}{n(\mathrm{ctx}_j) + |V| + 1}, \tag{3}$$

$$s_i = \frac{c_i}{100} + 0.5\,\mathrm{LM}(t_i) + \mathbb{1}\big[t_i \in \mathcal{L}\big], \tag{4}$$

where g_j is the j-th character trigram, ctx_j its two-character context, and $\mathcal{L}$ the lexicon. The hypothesis with the highest s_i is kept; its original c_i is returned as confidence.

d) Tier 3: abstention.: If neither reader yields text the field is empty. In the field-level configuration used for the case studies, a field whose fused confidence falls below the gate (Eq. 6) is additionally dropped rather than guessed. The default page-level path currently keeps the uncorrected reading and does not yet apply this gate; porting it is listed as future work.

D. Stage 3: lexicon correction

Exact fuzzy search over $\approx 3.8{\times}10^5$ names with Levenshtein distance [20] is too slow per crop. We instead index the lexicon once with character-n-gram TF-IDF [21] ($n \in [2,4]$, word-boundary padded) and retrieve the nearest entry by cosine similarity,

$$m^\star = \arg\max_{m \in \mathcal{L}} \frac{\mathbf{v}_q \cdot \mathbf{v}_m}{\|\mathbf{v}_q\|\,\|\mathbf{v}_m\|}, \tag{5}$$

which tolerates insertions, deletions and substitutions typical of OCR (e.g. `Amoxcilin`→`Amoxicillin`). A medicine line such as `Tab. Ozad 100 BC 7 AM` contains non-drug tokens, so the line is first split at digits and dosage keywords (`tab, cap, syr, inj, bd, od, tds, sos, hs, ac, pc, mg, ml, ...`) into candidate word sequences. For each sequence the longest prefix whose best cosine is ≥ 0.60 is substituted in place, leaving strengths and frequencies untouched. Text below threshold is returned

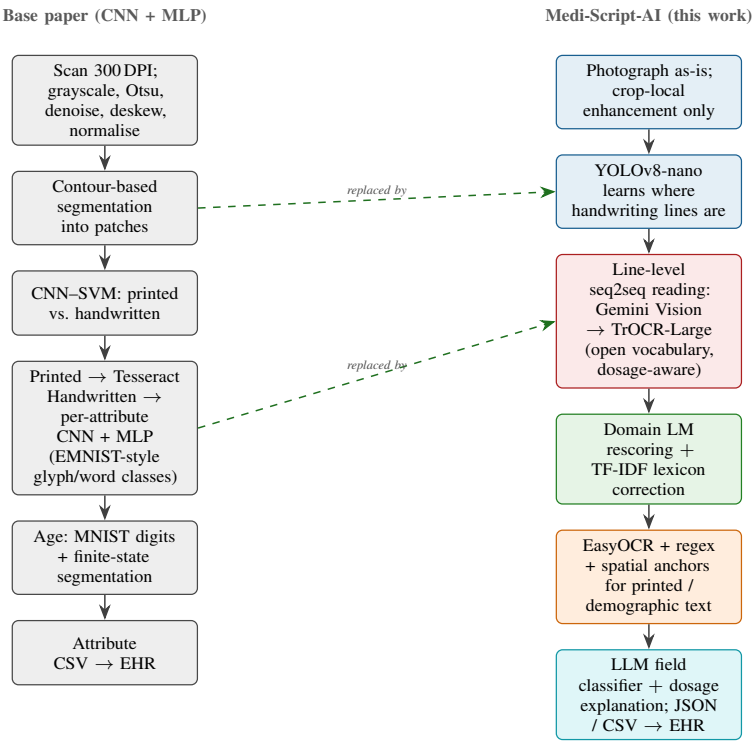


Fig. 3: Side-by-side pipelines: the base paper's CNN–MLP chain (left) and MEDI-SCRIPT-AI (right). Dashed arrows indicate which base-paper component each of our stages replaces.

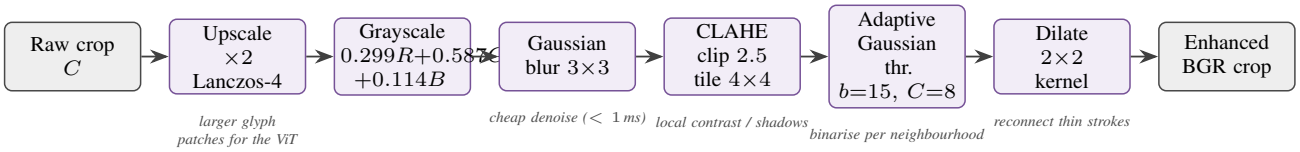


Fig. 4: Crop-enhancement chain (Stage 2a). Parameters are those used in the released code.

unchanged, which is what allows the gate to expose genuinely unreadable lines instead of inventing a drug.

An RapidFuzz [22] `WRatio` matcher with a first-letter index is used for the small field lexicons (gender, symptoms), and a character trie (`LexiconTrie`) provides $O(k)$ exact and prefix look-ups for the LM and for the optional TrOCR/EasyOCR reconciliation module (agreement if ratio > 85).

a) Symbol-aware repairs.: Two regular rewrites address systematic confusions: `1H1`→`1+1` (letter H read for the plus sign) and `400mg16mg`→`400mg/16mg` (dropped slash). The earlier unconditional substitution of $H, l, O \to +, 1, 0$ over the whole string corrupted drug names; the released version only rewrites tokens that look like dosage notation.

b) Confidence fusion.: The stage confidences—segmentation c_{seg}, recognition c_{rec} and validation c_{val} (90 if the lexicon changed the text, 55 if kept, 10 if no match)—are combined as

$$c = \min\Big(0.2\,c_{\text{seg}} + 0.3\,c_{\text{rec}} + 0.5\,c_{\text{val}},\ \ \min(c_{\text{seg}}, c_{\text{rec}}, c_{\text{val}}) + 15\Big), \tag{6}$$

so that one very weak stage cannot be hidden by two strong ones. In the field-level configuration, fields with $c < 15$, or a prescription field with $c_{\text{val}} < 50$, are dropped.

E. Stage 4: layout and demographic parsing

Everything outside the detected boxes is read by EasyOCR (CRAFT detector [15] and CRNN recogniser [5], [16]). Fragments with OCR confidence $< 15\%$ are discarded, and a fragment b is excluded when its intersection-over-area with any line box exceeds 0.1:

$$\text{IoA}(b, B) = \max_{b_j \in B} \frac{|b \cap b_j|}{|b|} > 0.1. \tag{7}$$

Remaining fragments are classified by ordered regular-expression rules plus position priors (Table III); normalised (x_c, y_c) coordinates implement the observation that hospital information sits in the top and bottom bands or the left margin.

a) Spatial anchor parsing.: Labels such as *Name*, *Age*, *Sex* and *Date* act as anchors (matched exactly or by fuzzy ratio ≥ 75 to survive `Name`→`Namc`). For an anchor a the value is the nearest fragment b to its right on the same line (Fig. 6),

$$b^\star = \underset{b:\ |c_y(b) - c_y(a)| < \delta,\ x_1(b) \geq x_2(a)}{\arg\min} \big(x_1(b) - x_2(a)\big), \tag{8}$$

with $\delta = 30$ px (15 to join a multi-word name, which ends at the *Sex* anchor; 45 for sex; 75 for date, where the best digit-and-separator candidate is preferred). Fall-backs recover

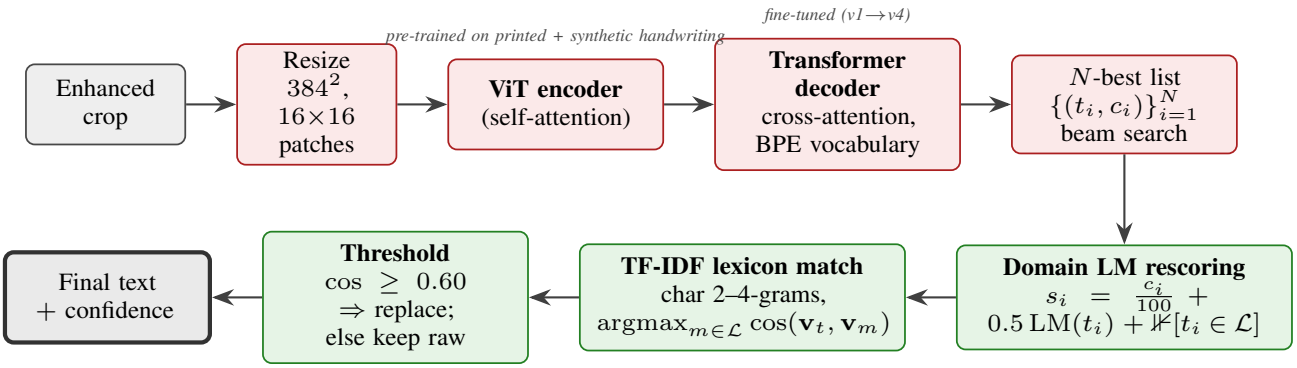


Fig. 5: Tier-2 recognition and Stage-3 verification: TrOCR N-best decoding, domain-LM rescoring (Eq. 4) and TF-IDF lexicon matching.

TABLE III: Ordered classification rules (first match wins).

Class	Rule
DoctorName	`dr.`, `md`, `mbbs`, `prof.`
PatientInfo	`name/namc/age/sex/gender/patient`; or `\d{1,3}` / `[mf]`, `\d+yrs`
Date	`date`, or `dd/mm/yy`; or a 6–10 char digit string in the top-right quadrant
Hospital / Clinic	$y_c < 0.25$ or $y_c > 0.8$ or $x_c < 0.25$, and keyword (`hospital`, `clinic`, `care`, `institute`, `...`) or ≥ 2 words
Other	fall-through

from missing anchors: a title (*Mr./Mrs./Mast.*) implies a name; the patterns `28/M` and `12 yrs / F` yield age and sex; date patterns tolerate alphanumeric OCR errors; a misread `47/F` ("4 yr/F") is repaired; the *Sex* value is re-read from a crop with TrOCR when EasyOCR fails; and `1717/24`→`17/7/24` corrects a slash read as "1".

F. Stage 5: constrained LLM post-processing

Two optional calls use a Gemini text model. The *other-field classifier* receives the residual `Other` string and returns JSON assigning fragments to patient name, age, sex, doctor, date or clinic; the result is applied *only to fields that are still empty* (clinic text is appended), and leftovers stay in `Other`. A doctor name must carry an explicit title, otherwise it is a patient name. The *dosage-advice generator* converts lines such as `T. Ambulax 1 BD` into "Take 1 twice a day (BD)". Both prompts forbid diagnosis or advice beyond translating abbreviations, and any API failure leaves the record unchanged.

G. End-to-end procedure

Algorithm 1 summarises inference.

Algorithm 1 MEDI-SCRIPT-AI inference

Require: image P; lexicon $\mathcal{L}$; thresholds θ_{det}=0.25, τ=0.60
$B \leftarrow \textsc{Yolo}(P, \theta_{\text{det}})$; $\mathcal{M} \leftarrow [\,]$
for $b \in B$ **do**
  $C \leftarrow \textsc{Enhance}(\textsc{Crop}(P, b, \text{pad}{=}5))$
  $t \leftarrow \textsc{Gemini}(C)$
  if $t = \varnothing$ **then** $t \leftarrow \textsc{Rescore}(\textsc{TrOcr}(C))$
  end if
  $t \leftarrow \textsc{LexiconCorrect}(t, \mathcal{L}, \tau)$; append t to $\mathcal{M}$
end for
$F \leftarrow \textsc{EasyOcr}(P)$; drop low-confidence and IoA > 0.1 fragments
classify each $f \in F$ (Table III); run anchor parser ⇒ patient, date
optional: $R \leftarrow \textsc{OtherClassifier}(R)$; advice $\leftarrow \textsc{Dosage}(\mathcal{M})$
return R

V. TRAINING

A. Detector

YOLOv8-nano was initialised from COCO weights and fine-tuned for 50 epochs at 640×640 on D2 (Tesla T4, about four minutes, 0.067 GPU-hours), with the library's default mosaic and colour augmentations and early stopping (patience 10 in the released training script). The checkpoint is stored at `models/yolo/yolo_real_world_best.pt`.

B. Recogniser lineage

TrOCR-Large was adapted in four documented steps (Fig. 7, Table IV). The reasoning was to *progressively unlock capacity while lowering the learning rate*: an initial run trains only the decoder so the encoder keeps its general stroke features; the cursive-specialisation run (v3) unfreezes the top four ViT layers, injects frequent drug names as single tokens (the embedding matrix is resized accordingly) to shorten decoding, and adds elastic and perspective distortion to emulate slant and stretch; the final increment (v4) teaches dosage syntax with synthetic data at a learning rate small enough to avoid catastrophic forgetting. Model selection uses the character error rate (CER) on a 15% validation split with early stopping.

VI. IMPLEMENTATION

The code base is a single installable Python package, `mediscript`, organised by pipeline stage (`detection`, `preprocessing`, `recognition`, `postprocessing`, `parsing`, `llm`, `evaluation`, `data_gen`, `training`). Heavy models are cached across requests, secrets are read from the environment, and all paths derive from one configuration module. A FastAPI service exposes `POST /analyze` and a static HTML front end

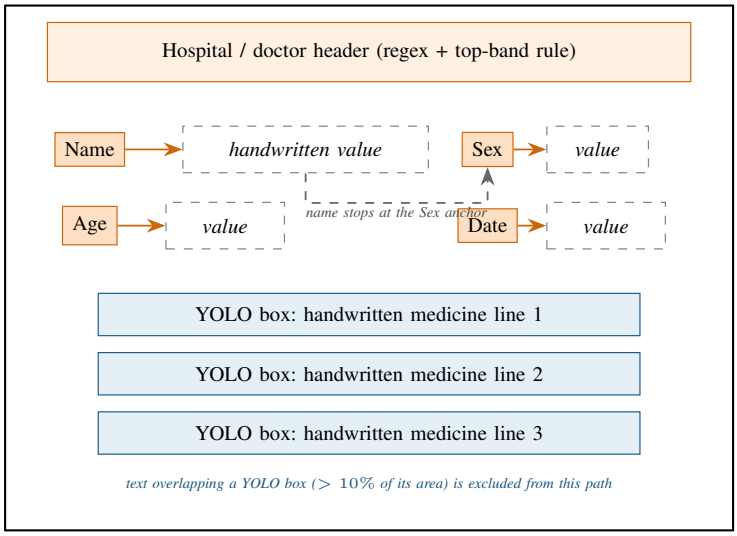


Fig. 6: Spatial anchor parsing on an abstract layout. Orange boxes are anchors found by EasyOCR; dashed boxes are the values located to their right. Blue boxes are YOLO regions handled by the other path.

TABLE IV: TrOCR fine-tuning configurations (from the released training scripts).

	v2	**v3**	**v4**
Trainable	decoder	top-4 ViT + decoder	decoder
Epochs (max)	5	20	5
Learning rate	5×10^{-5}	5×10^{-6}	1×10^{-6}
Batch (eff.)	8 ($\times$2)	4 ($\times$2)	4
Max label len.	64	128	128
Scheduler	linear, warm-up 50	cosine, 10% warm-up	default
Early-stop patience	3	5	2
Augmentation	rot., jitter, blur	+ perspective, elastic	perspective, rot.
Extra tokens	—	medical vocab.	(inherited)

TABLE V: Software components.

Function	**Component**
Line detection	Ultralytics YOLOv8-nano
Page OCR	EasyOCR (CRAFT + CRNN)
Handwriting OCR	TrOCR-Large (HF Transformers); Gemini 2.5 Flash
Fuzzy matching	scikit-learn TF-IDF; RapidFuzz
Image processing	OpenCV, Pillow
Serving	FastAPI, Uvicorn, HTML/JS
Training	PyTorch, HF `Seq2SeqTrainer`, Colab/Kaggle T4

TABLE VI: Handwritten-line detection (YOLOv8-nano, single ~~class,~~ 50 epochs).

Metric	**Value**
mAP50	86.1%
mAP50–95	46.2%
Precision	81.9%
Recall	80.6%
Training time (Tesla T4)	∼4 min (0.067 GPU-h)
Checkpoint size	6.2 MB
Inference (Tesla T4, per image)	2.5 ms

(drag-and-drop upload, structured result cards) calls it; a Dockerfile and a command-line entry point are provided. Gemini calls are optional: without a key the system degrades to the local TrOCR route, and LLM stages are skipped. Table V lists the main components.

VII. Results and Analysis

We report what was measured and, equally explicitly, what was not. Metrics for recognition are CER and word error rate

$$\mathrm{CER} = \frac{d_{\mathrm{Lev}}(\hat{t}, t)}{|t|}, \qquad \mathrm{WER} = \frac{d_{\mathrm{Lev}}(\hat{\mathbf{w}}, \mathbf{w})}{|\mathbf{w}|}, \tag{9}$$

with d_{Lev} the Levenshtein distance at character and word level; the released `evaluation` module also reports exact-match accuracy.

A. Field localisation

Table VI and Fig. 8 summarise the detector on the held-out portion of D2.

The gap between mAP50 (86.1%) and mAP50–95 (46.2%) is typical of handwriting regions: boxes are usually placed on the right line but with imprecise extent, because strokes bleed past nominal field boundaries and annotators drew boxes loosely. Recall trails precision, which we attribute to inconsistent annotation of overlapping or omitted lines; better curation rather than longer training is the promising lever. We have not benchmarked CPU-only inference on edge hardware, so the 2.5 ms figure should not be read as an edge latency.

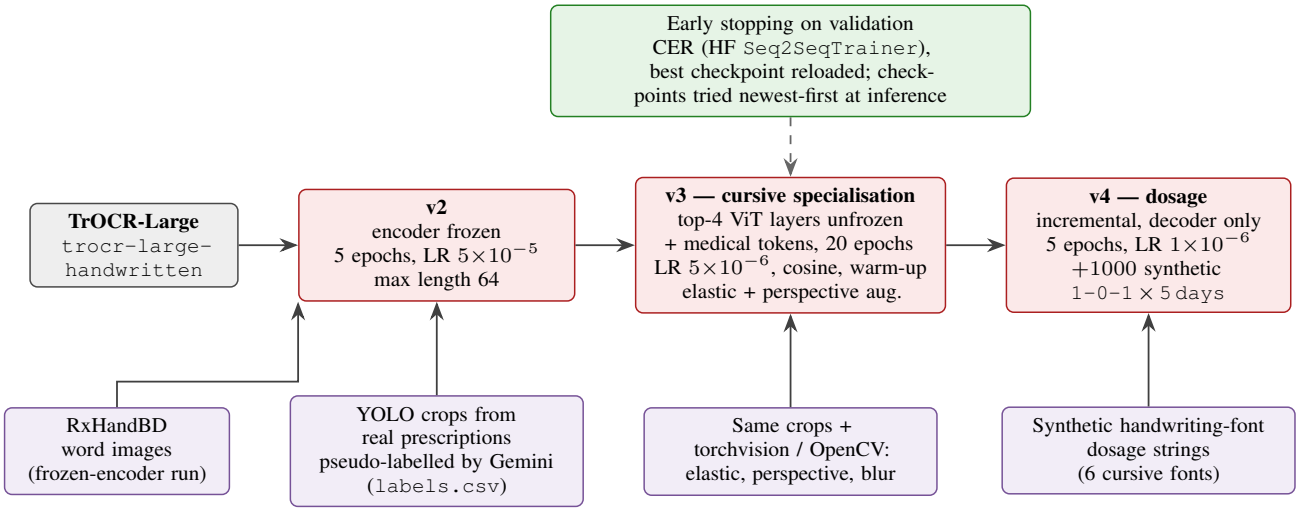


Fig. 7: TrOCR training lineage and the data feeding each stage.

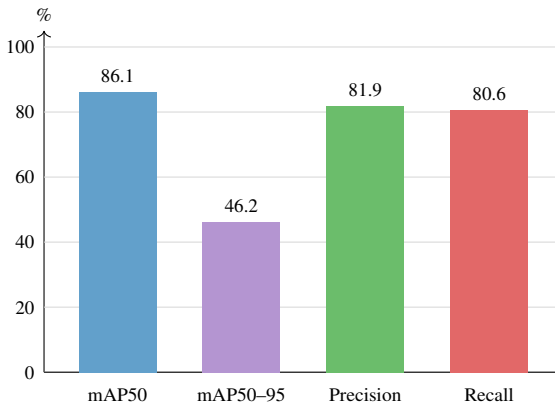
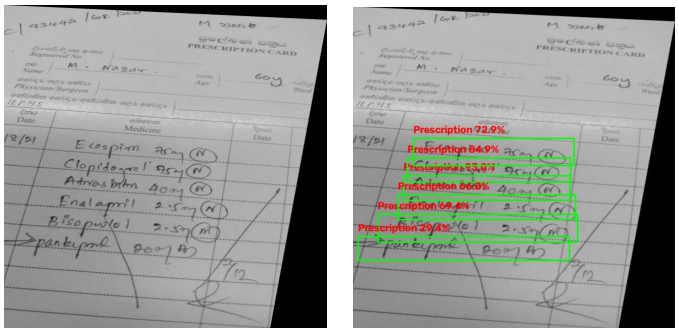


Fig. 8: Detector metrics from Table VI.



(a) Photograph (b) Stage-1 output

Fig. 9: Out-of-distribution test image (Case B). The detector boxes the handwritten medicine line tightly, localising the fault to recognition.

B. Qualitative end-to-end analysis

Early integration tests, run with the confidence-gated pipeline and the curated 1,142-entry lexicon, produced two instructive failures.

a) Case A: in-distribution image.: On a held-out image from D2, YOLO localised four medicine lines correctly. The then-untuned TrOCR returned strings such as `"wrapportingly ."` and `"pur-todac itself"` at 32.5–53.8% confidence—not near-misses of drug names. All four fell below the matching threshold, Stage 3 declined to force a correction, and the record's prescription field stayed *empty*. The system therefore failed *safely*, and the fault was isolated to recognition.

b) Case B: out-of-distribution image.: On a hospital prescription photographed independently of any training data (Fig. 9), the detector again boxed the medicine line, but the decoder returned a fluent, high-confidence name of an unrelated commercial product. This is the hallucination mode predicted by a general-language prior and the principal reason that domain LM rescoring and lexicon verification are mandatory components rather than refinements.

C. Baseline: end-to-end document transformer

A pretrained Donut checkpoint fine-tuned only on synthetic digitally rendered prescriptions [9], [23], run on a real photograph, emitted a repeating sequence of full stops irrespective of content—a complete collapse under the synthetic-to-real gap. This supports our decision to keep detection and recognition separate and to train recognisers on real or font-based *handwritten* images.

D. Provenance logging

An early run seemed to extract the patient's name correctly; inspection showed that a configuration fault had routed the image through a flat-OCR fall-back, and the name was read only because the pad pre-printed "Name:". Per-stage provenance logging (console logs report which recogniser handled each crop) was added, and all results in this paper were re-verified against path-level logs.

E. Comparison with the base study

Table VII lists the accuracies reported in the base paper. The paper states different figures in its abstract, results text and

TABLE VII: Accuracies reported by the base paper [1] (as stated; not reproduced by us and not comparable with Table VI).

Attribute	Abstract/figure	Results text
Name	91.2%	91.2%
Age	95.3%	94.5% (1 digit), 89.6% (multi)
Gender	93.5%	88.6%
Symptom	90.8%	96.81%
Medicine	92.6%	95.01%
Printed vs. handwritten	—	97.3% (confusion matrix, 1000 patches)

TABLE VIII: Evidence status per component.

Component	Evidence
YOLO detector	Quantitative (Table VI)
Lexicon gate	Qualitative (Case A; safe failure)
Domain LM, trie	Implemented; no ablation reported yet
TrOCR v3/v4	Trained; weights and a real-handwriting CER benchmark not yet released
Gemini tier	Implemented; no accuracy measured
Spatial parser	Implemented with regression-style fallbacks; no labelled benchmark
LLM stage	Implemented; prompt-constrained; not clinically validated

conclusion (for instance gender 93.5% vs. 88.6%, symptoms 90.8% vs. 96.81%, overall 90.20%), and they are measured on a proprietary corpus of more than 400,000 prescriptions with classification over fixed classes. Our detector metric is a different quantity (localisation) on a public set. **We therefore make no claim that MEDI-SCRIPT-AI is more accurate than the base system.** What the comparison supports is the design-level argument of Table I: capabilities that a closed-class classifier cannot provide—open-vocabulary lines, dosage notation, knowledge verification and abstention, layout-free parsing—and a public, re-runnable code and data pipeline.

F. Component-level summary

Table VIII states the evidence behind each component, to prevent over-reading of the results.

VIII. DISCUSSION

A. What is new relative to the base study

- **From classifiers to readers.** Replacing per-attribute CNN–MLP classifiers by a detector plus line-level sequence recognisers removes the closed-vocabulary ceiling and the need to segment glyphs.
- **Verification and abstention.** A large lexicon, a domain LM and, in the gated configuration, a multi-signal confidence rule (Eq. 6) turn recognition errors into empty fields rather than plausible wrong drugs.
- **Dosage and layout.** Synthetic dosage fine-tuning and a spatial parser handle `1-0-1`-style frequencies and free-form headers, which the base design does not model.
- **Graceful degradation.** Three recognition tiers and optional LLM stages allow the same code to run online (best quality) or offline (local weights only).
- **Reproducibility.** Public data, scripts, a package with accompanying unit tests, an API and a container image replace a proprietary corpus.

B. Limitations and threats to validity

1) **No end-to-end field accuracy.** We have not yet built a transcribed real-prescription benchmark; only localisation is quantified. Case studies are anecdotal.
2) **External dependence.** The best reading tier calls a commercial API, subject to latency (the free tier forces ≈ 4 s delays, giving 20–40 s per prescription), availability and, critically, patient-data privacy.
3) **Data scale and annotation noise.** Detection uses 357 images with loose boxes; D1/D2 include Roboflow augmentations; pseudo-labels from an LLM contain label noise.
4) **Scope.** English Latin script; one detector class; the multi-field detector of the original design remains future work. Some test images mix Bengali, which the OCR does not read.
5) **Heuristic fragility.** Anchor parsing assumes labels precede values left-to-right; unusual layouts will fail.
6) **Gate not in the default path.** The abstention gate was validated in the field-level configuration only; the default page-level path returns uncorrected text, so low-quality readings can appear in the output.
7) **Safety.** The system is a research prototype and must not be used for clinical decisions without human verification.

C. Future work

(i) Label 200–300 real crops and report CER/WER/exact match for each tier with ablations (no LM, no lexicon, no enhancement); (ii) expand the detector to the six-field schema and add a text-region classifier for printed vs. handwritten text, as in the base paper; (iii) distil the LLM tier into a local model to remove external dependence; (iv) calibrate confidences and add a human-review interface; (v) extend to multilingual scripts and to structured EHR (FHIR) export; (vi) benchmark CPU and edge latency.

IX. CONCLUSION

MEDI-SCRIPT-AI reframes prescription digitisation as *detect, read, verify*: a 6.2 MB detector localises handwritten lines (86.1% mAP50), a tiered recogniser reads them open-vocabulary, a 384,604-entry lexicon and a domain language model check the result (with an optional abstention gate), and a spatial parser plus constrained LLM complete the structured record. Compared with a recent CNN–MLP framework trained on a proprietary corpus, the design removes the closed-vocabulary limit, adds knowledge-based verification and an abstention mechanism, and handles dosage notation and free layouts, while remaining reproducible. Its main weakness is the absence of a quantitative end-to-end benchmark on real handwriting; closing that gap is the immediate next step.

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